

STATEMENT OF WORK

For

FAME in Jet Fuel Analyzers (x3)

24 March 2026

1. **Background:**

The Aerospace Fuels Laboratories at Wright-Patterson Air Force Base (AFPET/PTPLA), Vandenberg Space Force Base (AFPET/PTPLE) and Cape Canaveral Space Force Station (AFPET/PTPLH) provide quality surveillance and investigative analytical support of flight critical products & processes directly related to the warfighter and space launch operations on behalf of all Air Force and other DoD agencies, as well as commercial entities. AFPET has a vision to create a flexible, dynamic center of excellence that is responsive to customers and strategically positioned and focused on contingency and warfighter operations. We provide the warfighter and space launch activities specialized testing services and capabilities that include petroleum, propellant, cryogenic, chemical and gaseous product for all aerospace vehicles, systems, and equipment.

2. **Scope:**

AFPET/PTPLA, AFPET/PTPLE & AFPET/PTPLH have a requirement for three (Qty: 3) Fatty-Acid Methyl Ester (FAME) Analyzers to perform specification testing on Jet Fuel. These will need to be durable & robust units for routine specification and/or investigative testing. Units must conform to the requirements set forth in ASTM D7797-23, *Determination of the Fatty Acid Methyl Esters Content of Aviation Turbine Fuel Using Flow Analysis by FTIR*.

Note: Copies of ASTM Methods may be obtained by writing to ASTM International, 100 Barr Harbor Drive, PO Box C700, West Conshohocken, PA 19428-2959; or by telephone at 610-832-9255; or by e-mail at service@astm.org; or through the ASTM website at www.astm.org

3. **Minimum Technical Requirements:**

The following specifications list the minimum requirements for the FAME Analyzers:

- 3.1. Shall be three (3) instruments (with any strictly necessary add-ons) capable of automatically running ASTM D7797-23 and shall meet the following requirements:
 - 3.1.1. Electrically configured for standard 100-240V single phase, 50/60Hz electricity. Equipped with 3 wire electrical cord with 20 amp plug
 - 3.1.2. Unit must be a standalone unit able to sit and operate on a laboratory benchtop; Must weigh ≤ 100 lbs and have dimensions $\leq 24'' \times 24'' \times 24''$ (HxWxD)

- 3.1.3. Fully automatic sample analysis requiring no pre-sample preparation; No external gases required; Sample volume required $\leq 100\text{mL}$; Ability to analyze 3-4 samples per hour
- 3.1.4. Must be compatible for use in analysis of Jet Fuels (e.g. Jet A, JP-5, JP-8, F-76, etc.); Additional ability to run middle distillate fuels or residual fuels not required.
- 3.1.5. Detection range of at least 10 mg/kg to 150 mg/kg in Aviation Turbine Fuels and 20 mg/kg to 50% in Middle Distillate Fuels; Must comply with ASTM D7797-23 precision and bias requirements
- 3.1.6. Ability to operate in environmental conditions ranging from 10-35 °C and $\leq 80\%$ RH
- 3.1.7. Must detect all FAME components (range C8-C22), with peak IR absorbance at the carbonyl peak at approximately 1749 cm^{-1} , as specified in standards such as Specification D6751 and EN 14214
- 3.1.8. In accordance with ASTM D7797, sample will be flowed through a sorbent cartridge designed to retain FAMES; FAME content will be analyzed in a 2-mm effective path length flow-through cell before and after flowing through sorbent cartridge; Concentration of FAME in sample will be determined using differences of absorbance with a PLS-1 model
- 3.1.9. Self-cleaning unit requiring no cleaning solvents; Equipped with waste collection line/container
- 3.1.10. Equipped with Touch Display Screen to show operating conditions, test status and results. Unit will have memory to record and store test results for audit purposes
- 3.1.11. Equipped with USB 2.0+ ports for connectivity
- 3.1.12. Ability to connect to printer to physically print out test results
- 3.2. Package shall include supplies and verification materials to run ASTM D7797 upon receipt and install for each individual instrument: 3x1 Box of Low Lint Wipes, 3x11 50mL sample beakers, 3x10 + 3x100 (330 total) Sorbent cartridges with 5 μm filters, 3x1 100mL 30ppm verification material, and 3x1 100mL 100ppm verification material.
- 3.3. Shall include on-site installation and verification of performance each respective AFPET Laboratory location (WPAFB, VSFB, CCSFS)
- 3.4. Shall include up to 4 hours of training and test operation for up to 4 AFPET personnel at each location.
- 3.5. Offeror shall provide technical support (e.g. phone/e-mail support to help identify, troubleshoot and resolve hardware and software problems)
- 3.6. Shall include electronic copies of all manuals and instructions for instrument operation.

4. Qualifying Country Products:

“All products delivered to satisfy this requirement shall be manufactured in a qualifying country source in accordance with Defense Federal Acquisition Regulation Supplement (DFARS) 225.872. Products from other than qualifying countries will not be accepted, and any costs associated with products from other than qualifying countries, including shipping, etc., shall be at the contractor’s expense.”

5. Delivery, Installation, and Lead Time:

5.1. The Contractor shall deliver and install the products within 12 weeks after contract award to the following addresses (one instrument + respective accessories/supplies per each address):

SHIP TO DoDAAC: FP2070
Aerospace Fuels Laboratory (FP2070)
AFPET/ PTPLA
2430 C Street Bldg 70 Area B
Wright Patterson AFB OH 45433-7632
POC: Ms. Gina Claghorn -
gina.claghorn.1@us.af.mil
937-255-0739

SHIP TO DoDAAC: FP2075
Aerospace Fuels Laboratory (FP2075)
AFPET/ PTPLE
1747 Utah Ave Bldg 6670
Vandenberg AFB CA 93437-5220
POC: Mr. Ryan Carroll –
ryan.carroll.22@us.af.mil
805-606-5873

SHIP TO DoDAAC: FP2078
Aerospace Fuels Laboratory (FP2078)
AFPET/ PTPLH
15251 Scrub Jay Road Bldg 54800
Cape Canaveral SFS, FL 32920
POC: Mr. Geoffrey Giarmo –
geoffrey.giarmo@us.af.mil
321-853-5441

A delivery/installation Point of Contact (POC) will be identified in the schedule of the final awarded contract who will be responsible for coordinating delivery, installation, and training.

6. Warranty:

The Contractor shall include at least a one (1) year warranty for parts and services.

7. Invoice and Payment:

“The contractor shall be paid upon submission of a properly prepared invoice and after final inspection and acceptance has been made. A proper invoice shall cite the contractor’s name and address, invoice number, invoice date, order number, contract line-item number (CLIN), description of the supplies or service, unit of issue, unit price and extended price, terms of any prompt payment discount offered, and who to notify in the event the invoice is defective. The invoice shall be submitted to in Wide Area Work Flow (WAWF) in accordance with the terms of the final contract award”

8. Delivery Procedures – Commercial Vehicles:

8.1. The company shall properly ship the system to the Aerospace Fuels Laboratories at Wright-Patterson AFB OH, Vandenberg SFB CA, and Cape Canaveral SFS, FL. All shipping costs to include insurance will be included in contractor’s proposal. Great care using standard, approved packaging methods shall be taken.

8.2. All vehicles larger than a large pick-up truck are required to be inspected by the Space Force or Air Force Base Commercial Vehicle Delivery Gate (CVDG) prior to entering the installation. Vehicles to be inspected include, but are not limited to, the following:

1. Step van/panel truck
2. Tractor/trailer, box and flatbed containing cargo
3. Tanker trucks
4. Box trucks
5. Tour buses
6. Garbage/recycled waste trucks
7. Concrete trucks/mixers, dump trucks
8. Cranes, recreational vehicles, petroleum tanker

8.3. This inspection will be conducted at the CVDG of each respective base.

8.3.1. If the vehicle has the product inside (concrete and asphalt trucks) and timely delivery is necessary due to product deterioration it does not need to enter the CVDG. To bypass the CVDG, the contractor shall submit a list containing drivers’ names, social security numbers and the state in which the driver’s license is held for those drivers who will be entering the base. This shall be accomplished 24 hours prior to requested entry time. If the driver’s name is not on the list, he/she will not be allowed access to the installation through these gates and the base will not assume liability for denied access.

8.3.2. All irregular deliveries will be sent for inspection. All visitors, which include contractors and commercial vendors without proper background checks and contractor passes, will not be allowed.